

TranSlider: User-Controlled Augmented Content Presence Visualization in AR Collaboration

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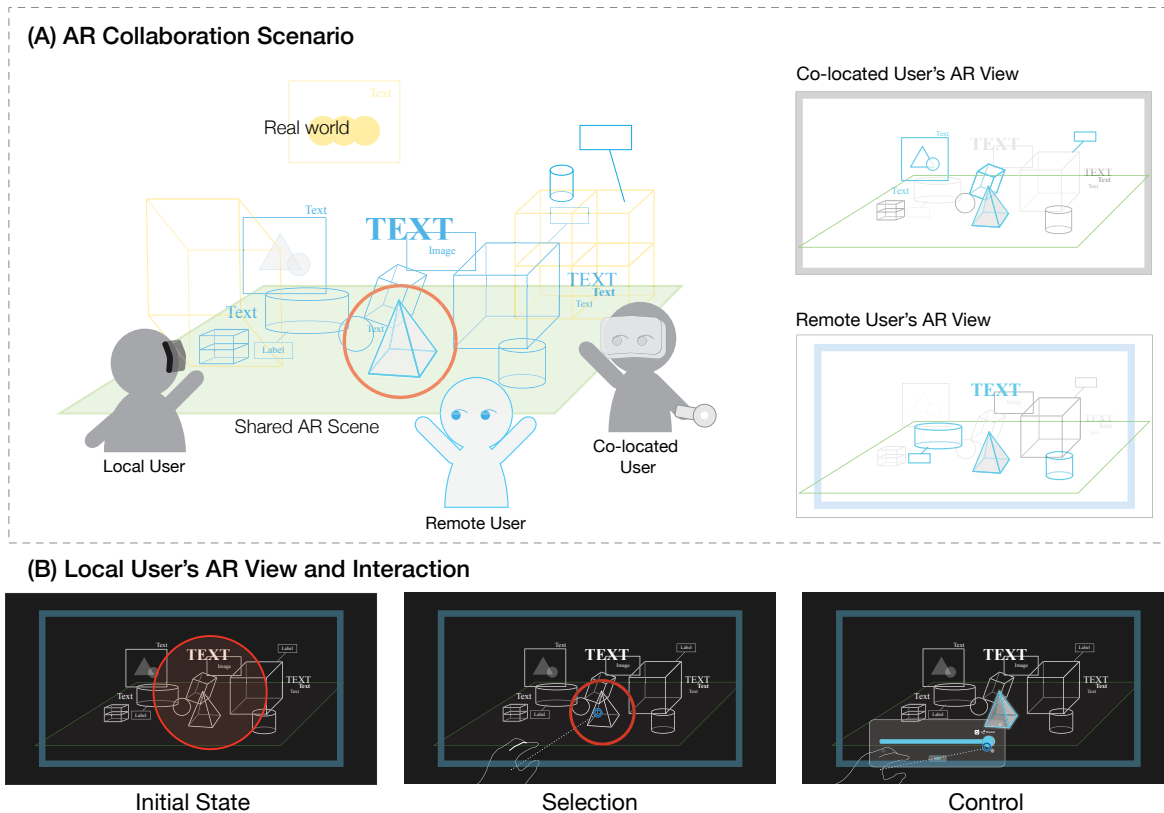


Figure 1: (A) It illustrates the main idea of the TranSlider interface: Users can control and more effectively achieve the augmented content's information in a collaborative AR scene. Participating users change each object's augmentation level and share their scene with other users; (B) Interaction process of the TranSlider: The initial state indicates the local user's limited cognitive area with a red circle. After selecting the desired target with a bare-hand gesture, the user can freely control its presence through the system.

ABSTRACT

We present a user interface method called *TranSlider*, designed for an AR collaboration with the concept of customized visualization of the user-selected augmented content. In AR, virtual content is augmented on top of the real world, so collaborating users are inevitably exposed to the diverse-overlapped information layers. Because the human recognition system has limited capacity, this overloaded information may burden the user to understand fully, and it is also challenging to manipulate among occluded objects. To solve this issue, we designed the *TranSlider* that allows users to control the presence of augmented content being transparent or highlighted

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based on the user interaction. The implemented prototype includes UI components, a manipulation function, and a network-connected collaboration module, and its applicability was verified through a preliminary interview. We expect that it would effectively support general AR collaborations and those with specific purposes, such as designing complex, hierarchical, or made up of many partial assets.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Collaborative interaction

1 INTRODUCTION

This study proposes the user interface methodology, which enables controlling the level of virtual content's augmentation based on the selection of users, for supporting an Augmented Reality (AR) collaboration. In the last decades, forward-looking technological developments in Head-Mounted Displays (HMDs), tracking devices, and 3D social applications have been introduced by commercial companies and research fields. This prodigious improvement makes it possible to collaborate between users regardless of their location.

In an AR collaboration, which augments the virtual information on the real environment, the local user can communicate with both the remote collaborator represented as a virtual avatar and the co-located collaborators. For instance, Mahmood et al. [5] showed the remote collaborative visualization system to share and discuss geospatial information with a distant partner's avatar. Luo et al. [4] and CloudBits [6] supported complex data sharing scenarios between co-located users in AR-based collaboration.

Likewise, the users can freely leverage real space information such as layout, furniture, or small movable objects, and virtual information such as text, images, 3D objects, or avatars for more effective collaboration. Therefore, during collaboration through the system, the users simultaneously face many information layers including the real world's physical environment, augmented information, and other users.

As presented in earlier works, it has a strong advantage in supporting social activities through various digital artifacts. However, some of this information can unconsciously be ignored or disturb users because the human cognitive system processes information limitedly depending on what they are currently focusing on. Also, it can be easily distracted and confused if they get overloaded with information at one time. Regarding the system and collaboration performance, this ignorance may lead to waste of system cost and cognitive burden; It will cause lower user concentration and communication effectiveness.

Therefore, we suggest a user interface method called TranSlider, which shows specific virtual information selected by the user as a transparent or highlighted representation through an input gesture. A swipe gesture is adopted to control the 3D virtual slider intuitively: the user moves their hand in the left or right direction to change the level of augmentation. Ahead of it, they first select which specific information will be dimmed or highlighted; This customized visualization can successfully utilize the user's main interest and present the necessary information for a better AR collaboration experience.

Interaction techniques and visualization methods, which handle complex, layered, and occluded information, have been developed for VR/AR. Yu et al. [9] designed controller interaction and Sidenmark et al. [8] used gaze input to select a fully occluded object in VR, and TunnelSlice [3] showed a subspace selection based on a free-hand interaction in AR. Furthermore, the following data visualization methods were investigated: automatically erase the information, make it transparent, change colors, or only reserve the basic structure of the object [1, 2, 7]. Unlike TranSlider, however, none of these interaction and visualization methods considered 3D collaborative context or user's main focus of interest, but only supposed the single-user usage scenario.

In other words, through the adjusted augmentation level of virtual content, it will be possible to achieve collaboration effectiveness by increasing user attention and understanding of information. Consequently, the suggested interface will be beneficial for the following aspects: First, for the users, it could lessen confusion and cognitive load from the information overload. At the same time, the users can focus more on the necessary information through the selection. Second, it can support more effective AR collaboration and improve communication efficiency. Also, the system resources may be saved because of the conditional rendering in some situations.

2 TRANSLIDER DESIGN AND IMPLEMENTATION

2.1 Main Concept and Interaction

Our TranSlider method has the main feature—controlling the augmentation level between transparent or highlighted—and is operated by the following steps: Initial state, Selection, and Control as explained in Figure 1(B). It also includes a collaboration-related feature of private and sharing mode and a multiple target control for better applicability. For a more intuitive utilization of TranSlider, it is also designed the main UI panel to visualize a controlling slider.

(1) **Initial State** At first, if the user selects none of the augmented content in the AR collaboration scene, the TranSlider interface is not activated; The main UI panel is not shown in the user interface but gets ready to take input.

(2) **Selection** The user can select the target object by posing a pinch gesture (as a click) along with the ray-casting method. Once the content is selected, the TranSlider control UI is popped up and is placed between the user and the selected object within the user's arm's length. When the user selects other targets, the prior one is deselected, and the augmentation level of the new object is ready to be adjusted by the TranSlider. To perform multiple selections, the user should click the desired objects sequentially, one by one, before adjusting the control UI value. If the user cancels the selection of all objects by clicking on an empty area in the 3D space, the TranSlider UI automatically disappears.

(3) **Control** Using the TranSlider, the user can control the augmentation level of AR content. For example, if the user changes the slider level to the left with the pinch and swipe gesture, the selected information gets dimmer, which means the transparency gradually increases. Conversely, if the user moves the slider to the right side, the target gets highlighted; the outline will shine. By adopting the familiar object manipulating gesture (pinch and swipe), the user does not need to get accustomed to the cumbersome operation.

Sharing for Collaboration Users can choose the selected target in private or shared modes for different accessibility. In the private mode, changes on the content will not be shared to their collaborators. Otherwise, in the sharing mode, all the visualizing effects will be shared to other users. This information-sharing mode can be utilized through an option button on the upside of the main UI (Figure 2).

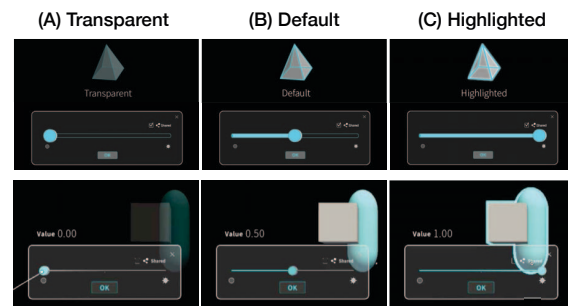


Figure 2: The three visualization modes of the TranSlider: (A) Transparent, (B) Default, and (C) Highlighted; Top row: main control UI; Bottom row: multiple-selected content's visualization implementation.

2.2 User Interface Design

As the main function of the TranSlider interface, the visualization changing method is mainly designed. Conceptually, the adopted transparent and highlighted mode is shown to users as illustrated in Figure 2. The object is always initially started in a default status (Fig. 2(B)). To show the selected information at least visible, even in the transparent mode, the lowest level of augmentation should be more than 0 percent for delivering its presence to the users (Fig. 2(A)). For the highlighted mode, we added an extra outline effect around the selected object so that the original object is preserved but emphasized by the outline. From the default status, if the user starts changing the slider towards a higher value, the outline firstly appears subtly. After the value increases, the outline material is also getting brighter (Fig. 2(C)).

The represented UI panel is the main component of the TranSlider interface, as shown in Figure 2. It consists of several graphical UI elements such as buttons, a checkbox, icons, and a back panel.

The main slider element is located in the center of the panel. The circular-shaped thumb in the middle is movable, so the user grabs and moves the circular handle; The effect goes to be translucent or highlights its outline. On each side of the slider, two icons indicate each of the transparent and highlighting modes. On the left top of the slider, a text field indicates the changing level in numeric value—default value of 0.5—for a more concise adjustment. The ‘OK’ button under the slider works to confirm the changing result on the selected object. For the sharing function, a checkbox is used to make it on and off: It is located on the top-right side of the panel to avoid unnecessary takeaways of the user’s attention. When the user checks the checkbox, the sharing mode is activated: If not, it remains as the private mode. On the right upper side of the panel, there is a panel closing button, which is indicated as an ‘X’ mark. The main UI is a movable panel so the user can freely grab and adjust it to the desired location in the 3D space.

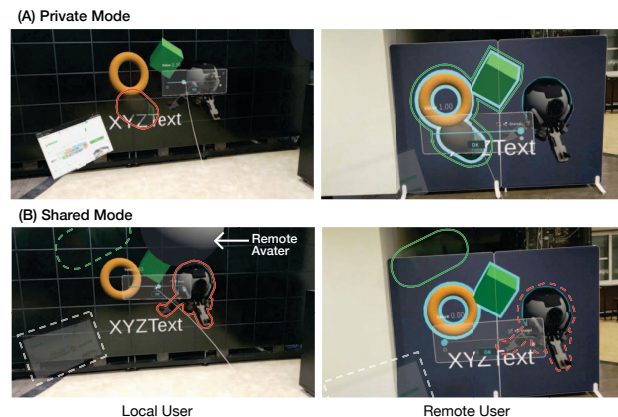


Figure 3: TranSlider also supports collaboration functions: (A) The private mode does not share any changing results (each object is shown differently); (B) The visual effect is shared and applied (as some objects are shown at the same level). (A red line: the local user’s interest; A green line: the remote user’s interest)

2.3 Setup and Implementation

For TranSlider interaction and graphical UI implementation, we adopted Microsoft HoloLens 2 as an AR HMD. To enable system control, basic manipulation, and visualize UI assets, we also used Mixed Reality Toolkit (MRTK 2.5.3) and AR Foundation (ver. 2.1.18). Our system supposed AR collaboration scenario, so Microsoft Azure Spatial Anchor—aligning location and synchronizing AR scene between users—and Photon Unity Networking Engine—sharing values of the augmented content—were also utilized. For the software, we worked on Unity 2019.4.26f LTS.

Visualization Effect As the primary function, each transparent and highlighted visualization effect was implemented by mapping the changed slider value to the relevant parameters in the selected object’s material information (see Fig. 2). To represent the transparent effect, we developed to make a change on an alpha level of the object’s color value. In the slider value of 0.5 (default), the alpha level is converted to 100%. Once the user decreases the slider value, the alpha level also relatively gets smaller. In our initial design, the lowest alpha level was set as 10%; however, this value was finally set as 5% based on the trial tests during implementation. If the slider handle moves to the left most, the target object’s transparency level is finally mapped to 5%. On the other hand, if the slider value increases, the highlight effect works by lighting the created outline. The slider value was also mapped to the brightness level of the outline material. In similar with transparent mode, the highest brightness level was also set based on the trial tests.

Networking and Sharing Each function of private and shared mode works based on the checkbox UI component. As illustrated in Figure 3(B), when the sharing checkbox is ticked, the subsequent changes in the user’s AR scene are shared with their collaborators (shared mode). On the other hand, the default checkbox status without a tick mark means private mode, which does not share the local user’s current scene with others (also see Fig. 3(A)). Because the TranSlider supports both remote and co-located AR collaboration, we developed a networking function through the Unity Photon networking module. Through the wireless network connection, the participating users can privately interact with the same augmented content within the AR scene or collaborate with others in sharing mode. As shown in Figure 3, for example, the default set of collaboration objects is the same, but the TranSlider results are differently applied and shown between collaborating users.

3 DISCUSSION

To preliminarily verify the suggested Slider UI concept, we asked about the system’s main idea, effectiveness, and applicability through an expert interview (Two female experts, ages 29 and 30; both have more than six years of experience in VR/AR HMD interaction). They agreed with the TranSlider’s needs and validity, so they showed positive responses: E1-“It will be useful in situations where the user wants to ignore other real and virtual objects while interacting with a specific and focused object.” Furthermore, it was also mentioned about the TranSlider’s future opportunity for supporting specific fields: E2-“It would be valuable under the scenarios if the virtual content has a hierarchical structure, consist of several small groups of parts, or have dependencies between objects, for example, designing a car model or utilizing an anatomical model of the human body.”

4 CONCLUSION

In this poster, we presented the user interface methodology to control the visualization level of virtual content according to the user’s preference to support better information achievement and AR-based collaboration. In AR collaboration, users are easily exposed to overloaded information across real and virtual environments. Our TranSlider interface displays certain virtual information controlled by the user input in a transparent or highlighted form to provide selective information acquirement under this issue. To utilize the proposed interface, we designed and implemented the TranSlider interface with the graphical UI, main interactions, and functions, including collaboration-related modules. From an expert interview, we conceptually confirmed the idea would support AR collaborations: it will deal with complex information and increase communication efficiency by greatly applying the user’s core interest. Thus, it may be more profitable for fields with strong 3D design aspects, such as layering units and hierarchical structures.

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